CorGPT: Coronary Angiography Imaging Analysis Using Large Medical Vision-Language Models

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Abstract. Recently, large vision-language models (VLMs) have demonstrated remarkable performance in multimodal tasks. However, their effectiveness in handling complex medical imaging remains limited. In this paper, we introduce CorGPT, an innovative multimodal medical dialogue model designed to analyze coronary angiography images and respond to open-ended questions. To bridge the gap between visual and language representations, we develop a bridge module that embeds visual features into the semantic space of a fine-tuned large language model (LLM). Additionally, we integrate a vector database (VecDB)-based autonomous retrieval mechanism to mitigate common challenges in medical AI, such as knowledge obsolescence and hallucinations. Experimental evaluations show that CorGPT outperforms baseline models, achieving significant improvements in ROUGE scores and higher consistency scores in GPT-based assessments. These results highlight its specialized capability and reliability in generating diagnostic summaries from coronary angiography images and supporting multi-turn medical dialogues. By fine-tuning on high-quality datasets, including but not limited to approximately 21,800 real-world cardiology patient-provider interaction transcripts, CorGPT achieves deep alignment between coronary angiography images and medical text, significantly enhancing diagnostic accuracy and multi-turn dialogue capability. This study provides new insights into the application of AI in medical imaging analysis.

Keywords: Large visual-language model · Coronary angiography images · Multimodal fusion · Image feature extraction · Retrieval-augmented generation

1 Introduction

Large vision-language models (VLMs) have made significant strides in bridging computer vision and natural language processing, achieving strong performance in tasks like image captioning [1], visual question answering [2], and medical image analysis [3]. These models require large-scale image-text datasets and benefit from domain-specific fine-tuning to enhance performance in specialized applications. Mini-GPT [4], for example, generates context-aware image descriptions but is limited in medical domains due to

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data scarcity and the need for precise domain knowledge (e.g., distinguishing "stenosis" from "occlusion"). To address these limitations in coronary angiography (CAG) analysis, we present CorGPT, a vision-language model fine-tuned from LLAVA using CAG radiology reports. CorGPT outperforms traditional summarization methods by generating concise summaries of key findings. It employs the MedClip [5] visual encoder and the Meta AI LLAMA-2 model [6] as its language backbone, along with a convolutional module and linear transformation layer for modality alignment. To further improve accuracy, CorGPT integrates a clinical knowledge retrieval mechanism via VecDB, mitigating issues such as outdated or hallucinated information [7].

Our key contributions include:

1. Domain-specific Dataset: A corpus of 21,800 doctor-patient dialogues was used to fine-tune LLAMA-2 for cardiovascular medical QA.
2. Enhanced Visual Encoding: The vision encoder was fine-tuned on ROCOv2 [8] and ARCADE [9] datasets to improve CAG feature representation.
3. Knowledge Retrieval Module: A curated set of 63,000 clinical guidelines supports external knowledge retrieval through VecDB.
4. Model Alignment: Integration of a convolutional module and transformation layer enables effective vision-language alignment for medical dialogue.

2 Related Work

In recent years, deep learning has achieved remarkable progress, particularly with convolutional neural networks (CNNs) and vision transformers (ViTs) making significant breakthroughs in medical image analysis [10]. These models have demonstrated excellent performance in tasks such as classification and segmentation; however, their adaptability to more complex radiological tasks remains limited. Concurrently, the rapid development of large language models (LLMs), such as ChatGPT, has led to unprecedented success in text generation, translation, and related applications [11].

In the medical domain, large language models (LLMs) are predominantly utilized in medical chatbot systems, offering personalized support and real-time assistance to both clinicians and patients. A representative example is ChatDoctor [12], which is trained on real-world doctor-patient dialogues and capable of incorporating external medical knowledge. This integration significantly enhances the model's response accuracy, particularly in understanding and addressing newly emerging medical terminology.

With the advancement of multimodal learning, there has been significant progress in the integration of computer vision (CV) and natural language processing (NLP), aiming to bridge the semantic gap between visual and textual information. Large vision-language models (LLVMs) jointly model multiple modalities, enabling machines to generate and interpret content that combines visual and linguistic features. Models such as Gemini [13] and Transfusion [14] adopt discrete encoders like VQ-VAE to tokenize visual inputs, which are then processed by multimodal transformers [15]. These systems employ fully connected attention mechanisms and causal attention masks to preserve sequence integrity. Benchmark models such as CLIP [16] and BLIP-2 [17] have demonstrated strong cross-modal performance in tasks including visual question answering, image captioning [1], and image generation [18]. Moreover, the LLAVA project integrates vision encoders with LLMs, yielding impressive results across various vision-language tasks [19].

Nevertheless, the effectiveness of these general-purpose multimodal models in medical image analysis remains uncertain. Multimodal models specifically tailored for healthcare are still scarce. Although open-source models like LLAVA-MED have been fine-tuned for the medical domain, their performance in interpreting coronary angiography (CAG) images requires further validation [20].

3 Method

3.1 Model Architecture

Figure 1 presents an overview of our CorGPT model. Given coronary angiography images, we align the visual features of images with the textual linguistic features, while designing a self-retrieval mechanism to enhance the accuracy of the model's language responses. Specifically, we use MedClip [5] as the visual encoder, Meta AI (LLAMA)-2 as our large language model (LLM) [6], and LanceDB [22] as the vector repository for our external knowledge. LanceDB is an open-source vector database specifically designed for AI applications. It supports vector search, full-text search, and SQL queries, optimizing the processing of multimodal data.

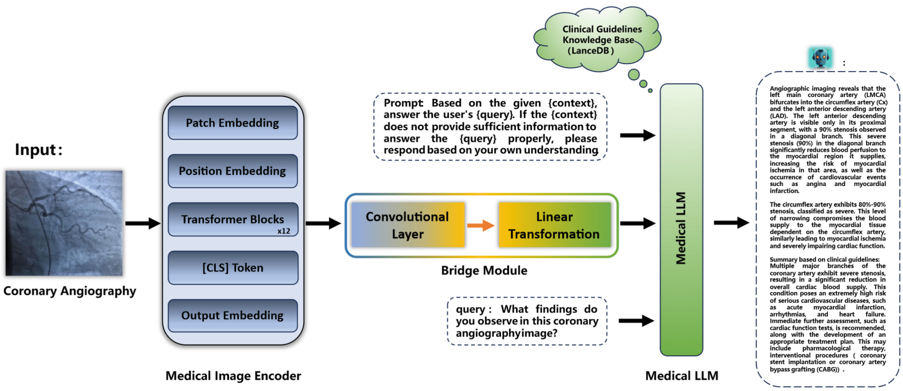


Fig. 1. Overview of CorGPT Framework: CAG images pass through five key components: **Image Encoder:** Extracts diagnostic features from coronary images. **Adaptive Convolution:** Enhances local features in critical regions. **Linear Transformation:** Aligns visual features with the medical LLM. **Guideline Retrieval (LanceDB):** Fetches relevant clinical references. **LLM Generation:** Combines retrieved context, user query, and image features to generate a detailed CAG summary.

Given a coronary angiography image $X \in I^{H \times W \times C}$, the vision encoder V_{image} encodes the image into embeddings. Subsequently, the raw embeddings are mapped to a dimension compatible with the language model embedding space via a linear projection layer, denoted as F_v:

$$F_v = G\big(V_{image}(X)\big) \tag{1}$$

Here, V_{image} is the vision encoder, and G is the projection layer.

Coronary angiography images have strong spatial correlations, with contextual relationships between slices and vascular structures. To capture these, we added an adaptive convolutional module with a convolutional layer, batch normalization (BN), and ReLU activation. This localized module enhances key regions' features (F_{image}) by applying convolution on the image modality features (F_v), aligning with the spatial nature of CAG images.

$$F_{image} = Conv(F_v) \tag{2}$$

Where $Conv$ represents the convolutional module.

To bridge the gap between image features and the language model's embedding space, we apply a trainable linear transformation layer N. It projects visual features (F_{image}) into language tokens (L_v), making them compatible with the language decoder (as shown in Fig. 2). This ensures both modalities share a common semantic space, enabling seamless interaction between image and text representations:

$$L_v = N\left(F_{image}\right) \tag{3}$$

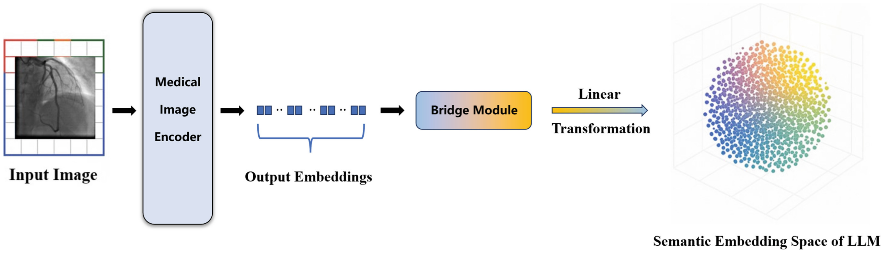


Fig. 2. The process of converting from image to semantic embedding.

To enhance the LLM with domain-specific knowledge, we introduce an autonomous retrieval mechanism using 63,000 cardiovascular clinical guidelines. These documents are embedded into semantic vectors via an embedding model [21] and stored in LanceDB [22] for efficient similarity-based retrieval. Retrieved content is combined with user queries and image-encoded features using a structured prompt, enabling the language model to generate accurate CAG summaries, denoted as output y_x.

$$y_x = M\left(C_q(D = \{d\})\right) \tag{4}$$

The designed prompt template is as follows:

"You are an expert in the medical field. Your task is to answer the user's $\{query\}$ based on the context $\{context\}$ and image feature $\{L_v\}$ provided below. If the $\{context\}$. in a does not adequately answer the $\{query\}$, you may answer based on your own understanding.

context: $\{context\}$; image feature: $\{L_v\}$; query: $\{query\}$."

CorGPT first lear visual-text encoding using the ROCOv2 [8] dataset, followed by training its bridging module with high-quality CAG image-text pairs from the ARCADE

[9] dataset, enhancing its radiological understanding and diagnostic generation capabilities. To strengthen domain expertise, we further fine-tune the model on a proprietary CQA dataset of real-world doctor-patient dialogues. During preprocessing, patient history information—identified as noise—was removed to improve model robustness.

3.2 Visual-Language Alignment

To ensure that the generated high-quality summaries align with the given CAG images, we fine-tuned the language model using a dialogue format similar to Meta AI (LLAMA)-2 [6], as shown below:

###Doctor: $\{L_v\}\{X_Q\}$ ###Assistant: $\{O_P\}$.

Here, $\{L_v\}$ represents the final visual representation produced by the linear transformation layer for the image $X \in I^{H\times W\times C}$. $\{X_Q\}$ is a randomly generated instruction (e.g., "What are the impressions and findings for the given coronary angiography image?"), and $\{O_P\}$ is the associated summary for the image X. In this manner, we compile image-text pairs accompanied by thorough and informative interactive summaries.

4 Designing High-Quality Data

ROCOv2 [8] is a multimodal dataset of 79,789 radiology images, each paired with medical concepts and captions from the PMC Open Access subset. The ARCADE [9] dataset contains 3,000 annotated XCA images for studying coronary artery disease (CAD). These images, covering the LCA and RCA, were validated by medical experts for accuracy. To generate concise medical summaries, we optimized CAG images from both datasets through expert annotations, excluding reports under 15 words and ensuring

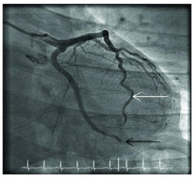
Coronary angiography Imaging	**Textual description of the imaging**
	Diagnostic coronary angiography shows a patent left anterior descending coronary artery (white arrow) and occlusion of the distal non-dominant left circumflex coronary artery (black arrow). The dominant right coronary artery (not shown) is unremarkable.
	The left main coronary artery is severely stenosed. The stenosis is located at the origin of the left main coronary artery, where blood flow is significantly reduced or interrupted due to the narrowing, potentially affecting the blood supply to downstream areas of the heart.

Fig. 3. We annotated and modified the ROCOv2 [8] and ARCAD [9] datasets based on medical expert annotation methods, making the textual descriptions more aligned with the imaging data and better suited for model training.

clinical accuracy. GPT-4 was used to remove patient history and predefined symbols, preserving the report's original meaning. The following (Fig. 3) is an example after preprocessing the ROCOv2 [8] and ARCADE [9] datasets. We curated a high-quality dataset of 21,800 de-identified doctor-patient dialogues on cardiovascular diseases from cardiology outpatient clinics, selected from an initial set of 30,000 after noise reduction. Figure 4 shows a sample preprocessed dialogue.

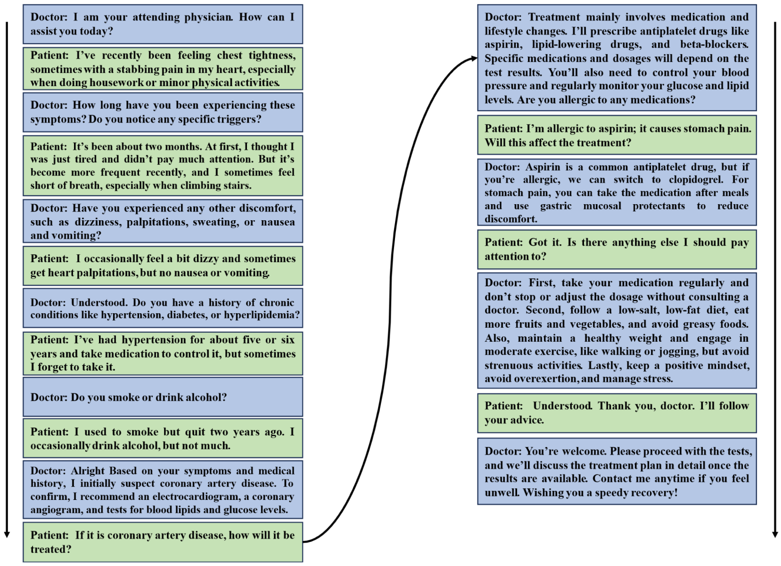


Fig. 4. Preprocessed Doctor-Patient Dialogue Example from Cardiovascular Disease Consultation

Based on the above processing, we obtained a filtered set of coronary angiography image-report pairs and a high-quality doctor-patient dialogue dataset. The image-report pairs and dialogue dataset include high-quality radiology image-text pairs from the ROCOv2 [8] and ARCADE [9] datasets, as well as 16,800 doctor-patient dialogues.

5 Experiments

5.1 Experimental Details

We convert CAG images into tokens through an image encoder, while question prompts are converted into textual tokens via a tokenizer. Both are processed by a causal language model to generate answers. Our training strategy consists of four stages to teach the model radiology-specific knowledge:

1. Image Encoder Pretraining: Fine-tuning a pretrained image encoder using a classification task to detect abnormalities in CAG images. Labels for "abnormal" and "normal" are assigned based on pathology findings using the ROCOv2 dataset.

2. Contrastive Learning: Using a CLIP-based approach, we train the image encoder and a text encoder to align medical images with corresponding text prompts (e.g., "arterial stenosis") from the ROCOv2 dataset. This enhances the encoder's ability to learn shared representations of images and text.
3. Aligning Visual and Language Representations: The image encoder is aligned with the language model via convolutional and linear transformation layers to enhance cross-modal consistency (as illustrated in Fig. 5). This stage is trained using the expert-annotated ARCADE dataset.
4. Instruction Fine-Tuning: We fine-tune the model with instruction-based training to improve response generation. This stage uses a private dataset of 21,800 doctor-patient dialogues from cardiology clinics, focusing on complex image interpretations. The model generates more informative reports and aligns with medical standards.

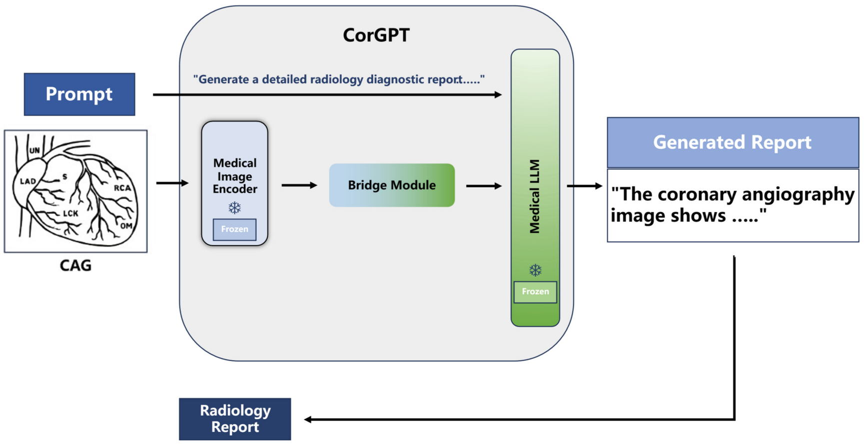


Fig. 5. Stage 3: Aligning Visual and Language Models for Diagnostic Report Generation.

5.2 Quantitative Measures

We evaluate our model using ROUGE scores, a widely adopted metric in natural language processing and text generation tasks [23]. ROUGE measures the overlap between generated and reference texts, focusing on the precision, recall, and F1-score of n-grams. Higher ROUGE scores indicate better accuracy, coherence, and overall textual quality. Additionally, we compare CorGPT with other multimodal models, including GPT-4 Vision and Gemini-Pro-Vision. Although these models are not fine-tuned for radiology-specific tasks, they demonstrate a certain level of capability in coronary angiography (CAG) image interpretation and medical question answering.

As summarized in Table 1, we present the key improvements of CorGPT over the baseline model MiniGPT-4 [4]. Quantitative evaluation using the ARCADE [9] test set shows that progressively integrating CorGPT's five core components into the MiniGPT-4 framework—culminating in the addition of the convolutional module—results in a 21% absolute increase in the ROUGE-1 score.

Figure 6 illustrates CorGPT's performance in detecting six critical pathological features, including stenosis, occlusion, irregular vessel walls, atherosclerosis, calcification, and in-stent restenosis. The model achieves an average F1 score of 0.59, outperforming GPT-4 Vision and Gemini-Pro-Vision, although slightly below the 0.66 F1 score reported by expert radiologists.

Table 1. Using Rouge scores on the ARCADE dataset.

Method	R-1	R-2	R-L
Baseline	0.1109	0.0117	0.0713
+ MedCLIP	0.1313	0.0204	0.0826
+ MedVicuna	0.1885	0.0447	0.1146
+Linear Transformation layer	0.2733	0.0661	0.1349
+ LanceDB	0.3098	0.0721	0.1496
+Convolutional Module	**0.3208**	**0.0808**	**0.1775**

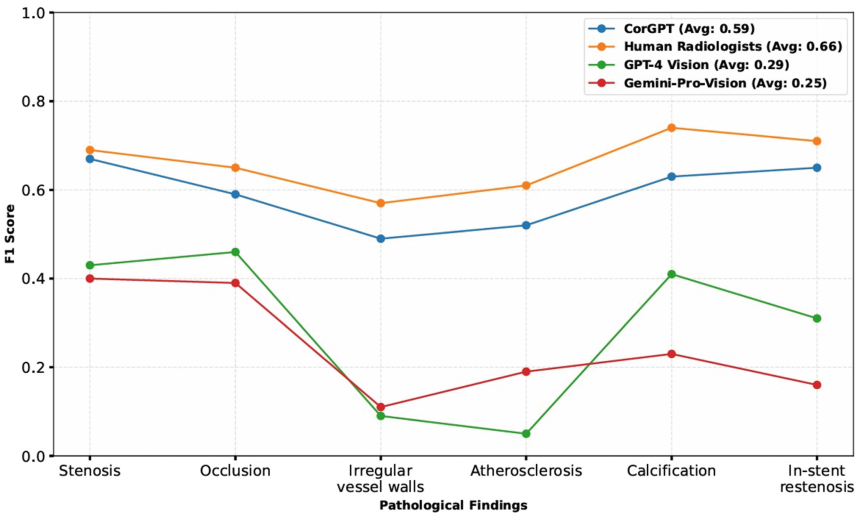


Fig. 6. The F1 scores of the trained vision encoder for classifying six key pathological labels in the test set.

5.3 The Generalization and Interpretability of the CorGPT

To assess CorGPT's generalization and interpretability, we constructed a real-world test set comprising 136 coronary angiography and 63 CTA images from patients aged 20–65, including various comorbidities. Following HAGNet [24] and CSLNet [25], we applied standard augmentations (e.g., rotation, scaling, brightness adjustment) and evaluated performance using n-gram recall scores. As shown in Table 2, CorGPT achieved a recall of 0.94 on angiography images, outperforming GPT-4 Vision. On CTA images—a different modality—CorGPT maintained a strong recall of 0.82 despite performance degradation, likely due to distributional shifts and higher spatial complexity, highlighting directions for future model refinement.

Table 2. N-gram Recall Comparison between CorGPT and GPT-4 Vision on Coronary Angiography Images across Patient Subgroups.

Patient Group	CorGPT	GPT-4 Vision
Hypertension only	0.94	0.83
Diabetes only	0.91	0.84
Hypertension + Diabetes	0.95	0.81
Male patients	0.96	0.83
Female patients	0.92	0.81
Average	**0.94**	**0.82**

Given the black-box nature of large language models, their interpretability remains limited. However, CorGPT incorporates a visual encoder and a contrastive learning-based training strategy, along with a bridging module that aligns visual and textual features in a shared spatial space. This design enables CorGPT to acquire diagnostic capabilities, including lesion localization in coronary angiography images (as shown in Figs. 7 and 8), thereby supporting clinical decision-making.

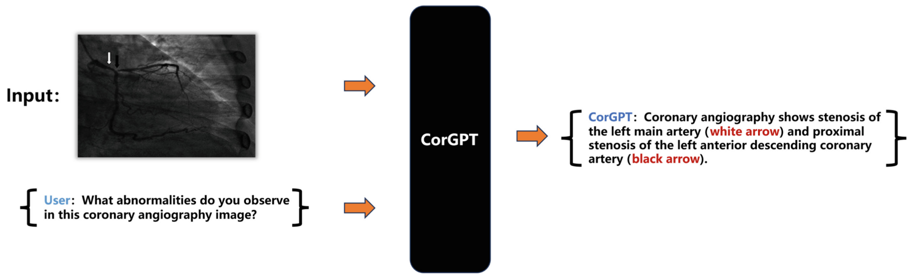


Fig. 7. CorGPT identifies the location of lesions in the coronary angiography image. Upon receiving the coronary angiography image and the user's inquiry, CorGPT analyzes both the image and the text, and the generated output indicates the location of the lesions in the image.

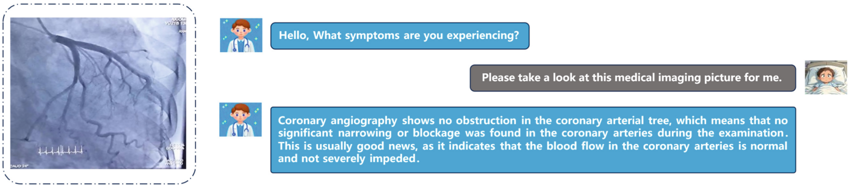


Fig. 8. CorGPT: Providing users with comprehensive yet concise discoveries and insights. Our CorGPT offers detailed imaging summaries.

6 Conclusion

In summary, we propose CorGPT, a multimodal medical vision-language model for coronary angiography (CAG) analysis. By aligning visual-text representations and integrating autonomous knowledge retrieval, CorGPT significantly improves diagnostic accuracy and response quality over baseline models. Nonetheless, limitations remain. Its generalization to rare cases is restricted by dataset coverage; high computational costs hinder deployment; and potential biases in clinical guidelines may affect output reliability. Future work will focus on dataset expansion, real-time knowledge updates, and architectural optimization to enhance robustness, accuracy, and efficiency.

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